

Aarush Agarwal

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EDUCATION

Carnegie Mellon University

Bachelor of Science in Artificial Intelligence & Computer Science

Expected Graduation, December 2027

3.78 QPA, Dean's List High Honors

Honors: USACO Gold, 6× Hackathon Winner (Claude, Harvard, Berkeley, 3× CMU)

EXPERIENCE

Shopify | Machine Learning Engineer Intern, Search Relevance

Jun 2026 - Present

- Designed a merchant-aware auxiliary training objective that improved a core merchant-search relevance metric by 10%, prioritizing first-party catalogs over third-party resellers for merchant-intent queries. Enabled the team to incorporate high-quality clicks as secondary supervision alongside relevance labels.
- Fine-tuned, distilled, and deployed Liquid AI LLM2.5 re-rankers on a custom Triton serving stack, cutting p99 inference latency by 5% (20 → 19 ms).
- Identified data/supervision mismatches through win/loss analysis of production rerankers vs. LLM reference judgments, establishing the need for new training data.
- Developing a unified Storefront/Shop/Catalog query-rewriting model (+0.5 NDCG) and improving non-English rewrites.

Shopify | Machine Learning Engineer Intern, Buyer Fraud

May 2025 - Aug 2025

- Co-filed a patent** on a distributed multi-agent system that decomposes tasks via a Neo4j dependency graph and executes subtasks across specialized agents in parallel.
- Improved buyer-fraud detection accuracy by 3% and reduced training iteration time by 70% through dimensionality reduction, importance-based feature pruning, and BigQuery/Dataflow + Vertex AI pipeline rebuilds.

CMU Language Technologies Institute | ML Researcher

Jan 2026 - Present

- Researching adaptive multimodal MoE architectures with Prof. Chenyan Xiong, targeting capacity expansion for OOD vision-language workloads while mitigating text-reasoning degradation during continual pretraining and fine-tuning.
- Building an autonomous Auto-MoE experimentation framework where specialist agents synthesize bounded architecture, routing, curriculum, and adapter interventions, scoring variants with multimodal loss proxies, router-health guardrails, and lm-eval reasoning benchmarks against baselines.

CMU Cosmology Lab & CERN | CUDA Researcher

Aug 2024 - Oct 2025

- First-authored FastGraph, a GPU-resident exact k-nearest-neighbor engine for geometric deep learning, delivering up to **41× speedups over FAISS-GPU**. Developed PCA-subspace spatial pruning that reduces the search space while evaluating distances in the original coordinates, preserving exact neighbors.
- Engineered compile-time-specialized C++/CUDA kernels with 2–5D spatial binning, hypercube neighborhood traversal, register-cached distance evaluation, custom backward gradients, and PyTorch/GravNet integration for differentiable graph-learning workflows.

PROJECTS

AutoReflex | Hardware & ML

Mar 2026

- 3rd Place at HackberryPi** for AutoReflex, a hardware-in-the-loop control system combining 100 FPS global shutter vision on a Jetson Orin Nano, UDP coordinate streaming, and a 1 kHz Raspberry Pi PID loop to drive a servo, solenoid, and TENS actuation in approximately 30 ms end to end.

Haven | Backend

Oct 2025

- Won 1st Place in the YC, LiveKit, and Fetch.AI tracks and Best Startup at CalHacks** for a patient-care platform; architected clinical-agent infrastructure, an rPPG heart-rate monitoring pipeline (OpenCV, FastICA), and real-time voice intake (LiveKit, Groq).

SKILLS

Languages Python, C++, CUDA, SQL, C, Swift

Software PyTorch, Triton, TensorFlow, JAX, OpenCV, GCP, Slurm, PostgreSQL, Neo4j

ML Systems Model Distillation, Distributed Training, Mixture-of-Experts, Computer Vision, Search Ranking